

Exact Equations: When Two Puzzle Pieces Fit

A ground-up companion to Lecture 4 — what makes a differential equation "exact," the test that tells you, the five-step method that solves it, and the integrating-factor trick that rescues equations that fail the test.

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The Big Idea: Reverse-Engineering a Hill

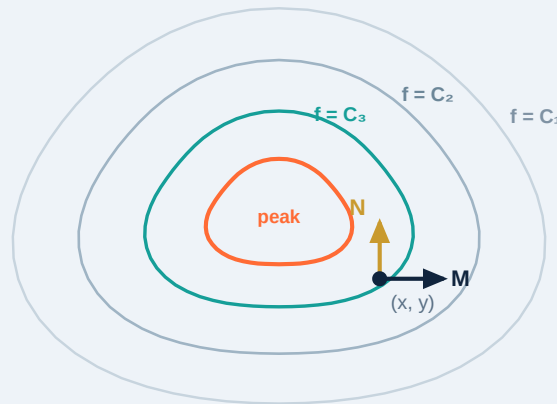
TL;DR

An exact equation $M dx + N dy = 0$ is secretly just the "flattened" derivative of some single function $f(x, y)$. Solving it means reconstructing that original f from its two pieces.

So far, every DE you've solved involved one variable changing with respect to another: dy/dx . This lecture introduces equations written as $M(x, y) dx + N(x, y) dy = 0$ — two pieces, one attached to a tiny step in x , the other to a tiny step in y .

THE ANALOGY

Picture a hill described by some function $f(x, y)$ — the height at every point on a map. If you take one tiny step east, the height changes by roughly $M dx$. If you take one tiny step north, it changes by roughly $N dy$. An **exact equation** is what you get when someone hands you those two slope-pieces, M and N , and asks: "can you figure out which hill these came from?" Sometimes the answer is yes — the two pieces are perfectly consistent with a single smooth hill. Sometimes the answer is no — the "slopes" you were handed contradict each other and could never have come from one real hill.



Each ring is a level curve $f(x, y) = C$ — a path of constant height on the hill. M and N are the local east/west and north/south slopes at a point.

That's exactly what "solving" an exact equation means: not finding a number, but reconstructing the hill $f(x, y)$ itself. Once you have it, the solution to the DE is simply $f(x, y) = C$ — you're walking along one particular level curve, one fixed "elevation," and C just says which one.

1 The Exactness Test: Checking the Story Is Consistent

TL;DR

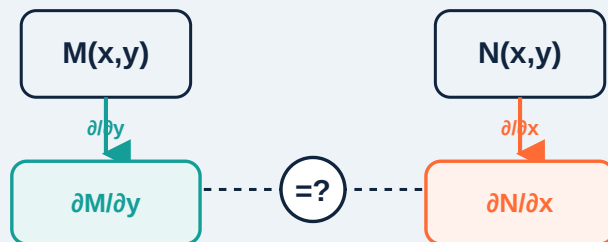
Before hunting for the hill, do a quick consistency check: take $\partial M / \partial y$ and $\partial N / \partial x$. If they match, a hill really does exist. If they don't, no such hill exists — at least not yet.

Formally: $M(x, y) dx + N(x, y) dy$ is called an **exact differential** if it equals df for some function f . The DE $M dx + N dy = 0$ is then called an **exact equation**. The theorem below is the practical test — you never have to guess:

THEOREM — CRITERION FOR EXACTNESS

Let $M(x, y)$ and $N(x, y)$ be continuous with continuous first partial derivatives on a rectangular region. Then $M dx + N dy$ is an exact differential if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$



Take the cross-derivatives of M and N and compare — if they agree, the equation is exact.

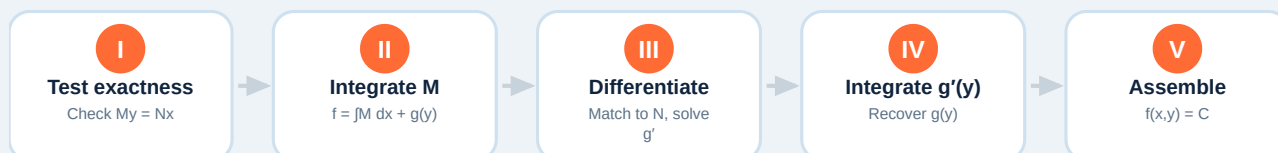
WHY THIS TEST ACTUALLY WORKS

If $f(x, y)$ is a genuinely smooth hill, a basic fact from calculus (mixed partial derivatives) guarantees that differentiating f_x with respect to y gives the *same* result as differentiating f_y with respect to x . Since $M = f_x$ and $N = f_y$ for a real hill, that fact becomes exactly $M_y = N_x$. So the test isn't a trick — it's just this "order doesn't matter" rule, dressed up.

2 The Five-Step Method: Rebuilding the Hill

TL;DR

Once you know the hill exists, rebuild it piece by piece: integrate M to get most of f , then use N to patch in whatever the first integration missed.



STEP I Confirm $\partial M / \partial y = \partial N / \partial x$. If they match, a function $f(x, y)$ exists with $f_x = M$ and $f_y = N$.

STEP II Integrate M with respect to x (treating y as a constant): $f(x, y) = \int M dx + g(y)$. The " $+g(y)$ " is a stand-in for anything that vanished when you differentiated with respect to x — exactly like the " $+C$ " in ordinary integration, except here the missing piece can depend on y .

STEP III Differentiate that result with respect to y , and set it equal to N . Solve for $g'(y)$:

$$g'(y) = N(x, y) - \frac{\partial}{\partial y} \int M(x, y) dx$$

STEP IV Integrate $g'(y)$ with respect to y to recover $g(y)$ itself.

STEP V Substitute $g(y)$ back in. The implicit solution is $f(x, y) = C$.

A SANITY CHECK WORTH KNOWING

After step III, $g'(y)$ should come out depending on y *only* — every x should cancel out. If it doesn't, that's a sign of an arithmetic slip (or that the equation wasn't actually exact) — always double check the exactness test from step I first.

3 Worked Example 1: A Clean, Short Case

PROBLEM

Solve $2xy \, dx + (x^2 - 1) \, dy = 0$.

STEP I $M = 2xy$, $N = x^2 - 1$. Then $M_y = 2x$ and $N_x = 2x$. They match — **exact**.

STEP II $f = \int 2xy \, dx + g(y) = x^2y + g(y)$

STEP III $f_y = x^2 + g'(y)$, and this must equal $N = x^2 - 1$, so $g'(y) = -1$.

STEP IV $g(y) = -y$ (absorbing the constant into C at the end).

STEP V $f(x, y) = x^2y - y$.

SOLUTION

$$x^2y - y = C \iff y(x^2 - 1) = C$$

4 Worked Example 2: An IVP in Disguise

PROBLEM

Solve $\frac{dy}{dx} = \frac{xy^2 - \cos x \sin x}{y(1 - x^2)}$, subject to $y(0) = 2$.

This one doesn't arrive in $M \, dx + N \, dy = 0$ form — it's disguised as an ordinary dy/dx . The first move is always to **cross-multiply and rearrange** before anything else:

$$y(1 - x^2) dy = (xy^2 - \cos x \sin x) dx \implies (xy^2 - \cos x \sin x) dx + (x^2 y - y) dy = 0$$

STEP I $M = xy^2 - \cos x \sin x$, $N = x^2 y - y$. Then $M_y = 2xy$ and $N_x = 2xy$. They match — **exact**.

STEP II $f = \int (xy^2 - \cos x \sin x) dx + g(y) = \frac{x^2 y^2}{2} - \frac{\sin^2 x}{2} + g(y)$
 (using the substitution $u = \sin x$ for $\int \cos x \sin x dx = \frac{1}{2} \sin^2 x$)

STEP III $f_y = x^2 y + g'(y)$, matched to $N = x^2 y - y$, gives $g'(y) = -y$.

STEP IV $g(y) = -\frac{y^2}{2}$

STEP V $f(x, y) = \frac{x^2 y^2}{2} - \frac{\sin^2 x}{2} - \frac{y^2}{2} = C_1$, or (doubling both sides) $x^2 y^2 - \sin^2 x - y^2 = C$

NOW USE THE INITIAL CONDITION

Plug in $x = 0$, $y = 2$: $0 - 0 - 4 = C \implies C = -4$.

PARTICULAR SOLUTION

$$x^2 y^2 - \sin^2 x - y^2 = -4 \iff y^2(1 - x^2) + \sin^2 x = 4$$

THE TAKEAWAY

Whenever you see a bare $dy/dx = (\text{stuff})$, don't assume it's not an exact-equation candidate. Cross-multiply into $M dx + N dy = 0$ form first — exactness might be hiding in plain sight.

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When It Fails: Rescuing a Non-Exact Equation

TL;DR

If the exactness test fails, all is not lost. Sometimes multiplying the whole equation by a clever factor — an **integrating factor** — repairs the mismatch and makes it exact.

THE ANALOGY

Think of $M_y \neq N_x$ as two mismatched puzzle pieces — close, but the edges don't quite line up. An integrating factor is like a tool that reshapes *both* pieces by the same amount (multiplying the whole equation by some function μ) until the edges suddenly fit. It doesn't change what curve the solution traces out — it just changes the equation into a form the exactness test can recognize.

PROBLEM

Solve $xy \, dx + (2x^2 + 3y^2 - 20) \, dy = 0$.

CHECK $M = xy$, $N = 2x^2 + 3y^2 - 20$. Then $M_y = x$ but $N_x = 4x$. **Not exact** — they don't match.

When the test fails, check whether one of these two combinations simplifies to a function of a single variable:

if $\frac{M_y - N_x}{N} = h(x)$ only: $\mu(x) = e^{\int h(x) \, dx}$ if $\frac{N_x - M_y}{M} = h(y)$ only: $\mu(y) = e^{\int h(y) \, dy}$

TRY $\frac{M_y - N_x}{N} = \frac{x - 4x}{2x^2 + 3y^2 - 20} = \frac{-3x}{2x^2 + 3y^2 - 20}$ — still has both x and y . **No good.**

TRY $\frac{N_x - M_y}{M} = \frac{4x - x}{xy} = \frac{3x}{xy} = \frac{3}{y}$ — a function of y alone. **This works.**

BUILD THE INTEGRATING FACTOR

$$\mu(y) = e^{\int (3/y) \, dy} = e^{3 \ln y} = y^3$$

Multiply the entire original equation by y^3 :

$$xy^4 \, dx + (2x^2y^3 + 3y^5 - 20y^3) \, dy = 0$$

RE-CHECK Now $M' = xy^4$, $N' = 2x^2y^3 + 3y^5 - 20y^3$. $M'_y = 4xy^3$ and $N'_x = 4xy^3$ — **they match now**. Exact, at last.

STEP II $f = \int xy^4 \, dx + g(y) = \frac{x^2y^4}{2} + g(y)$

STEP III $f_y = 2x^2y^3 + g'(y) = N' \Rightarrow g'(y) = 3y^5 - 20y^3$

STEP IV $g(y) = \frac{y^6}{2} - 5y^4$

STEP V $f(x, y) = \frac{x^2 y^4}{2} + \frac{y^6}{2} - 5y^4$

SOLUTION

$$x^2 y^4 + y^6 - 10y^4 = C \iff y^4(x^2 + y^2 - 10) = C$$

WHY MULTIPLYING DOESN'T CHEAT

Multiplying an equation by $\mu(y)$ doesn't change which curves $y(x)$ satisfy it (as long as $\mu \neq 0$) — it only changes the algebraic "packaging" of M and N so the exactness test can see the underlying hill that was there all along.



Tying It All Together

An exact equation $M dx + N dy = 0$ is just the flattened derivative of a hidden function $f(x, y)$ — the "hill" whose slopes in the x - and y -directions are exactly M and N . The test $M_y = N_x$ checks whether such a hill could possibly exist, using nothing more exotic than the fact that mixed partial derivatives don't care which order you take them in.

When the test passes, the five-step method rebuilds f piece by piece — integrate M , patch the leftover with N , done. When it fails, the equation isn't necessarily unsolvable this way — an integrating factor can sometimes reshape M and N until they agree, without ever changing the actual solution curve underneath.

A plain-English companion to: ENGR 213, Lecture 4 (Sept 18, 2026), Prof. A. Haghighat M., Concordia University — reference text: Zill, Advanced Engineering Mathematics (7th ed.), Section 2.4. Worked solutions to Examples 1–3 and the analogies (hill/level-curves, puzzle-piece integrating factor) are original, built to fill in and illustrate the same lecture material.